Helpful Notes for Solving Cryptography MCQ Exam

Based on the lecture slides, here are key concepts and memory aids to help you solve similar questions:

Lecture 01 - Course Overview & Key Concepts

1. CIA Triad (The Foundation)

- **C**onfidentiality - Privacy, secrecy (achieved by **Cryptography**)
- **I**ntegrity - No tampering (achieved by **Hashing**)
- **A**vailability - Systems are accessible

2. Security Layers (Memorize which protocol works at which layer)

- **Application Layer** → PGP (Pretty Good Privacy) - for Email security
- **Transport Layer** → SSL/TLS - for securing TCP segments
- **Network Layer** → IPSec (via VPN) - for securing IP datagrams

3. Types of Attacks

Passive Attacks (Eavesdropping - Hard to detect, easy to prevent)

- Release of message contents - Reading the actual message
- Traffic analysis - Observing patterns, who talks to whom, timing

Active Attacks (Modification - Easy to detect, hard to prevent)

- **Masquerade** - Impersonating someone else
- **Replay** - Capturing and resending valid data later
- **Modification** - Changing the content of messages
- **Denial of Service** - Flooding to make service unavailable

4. Cryptography Basics

- **Plaintext** → Original message
- **Ciphertext** → Encrypted message
- **Cipher** → Algorithm for transformation
- **Key** → Secret info used in cipher
- **Encipher/Encrypt** → Plaintext → Ciphertext
- **Decipher/Decrypt** → Ciphertext → Plaintext
- **Cryptography** → Study of encryption principles
- **Cryptanalysis** → Study of breaking codes without key
- **Cryptology** → Both cryptography + cryptanalysis combined

5. Key Fact for Exams

Algorithms are known; only keys are secret - This appears often!

6. Symmetric vs Asymmetric Cryptography

Feature	Symmetric	Asymmetric
Keys	One shared key	Public + Private key
Use case	Long messages	Short messages, key exchange
Speed	Fast	Slow

7. WiFi Security Evolution

Old & Insecure → New & Secure
WEP (RC4 - Deprecated) → WPA (TKIP - Deprecated) → **WPA2 (AES) → WPA3 (Most Secure)**

Lecture 02 - Traditional Ciphers & Cryptanalysis

1. Substitution Cipher

- Replaces **one symbol with another**
- **Monoalphabetic** - Each letter maps to ONE fixed substitute
  - Example: HELLO → KHOOR (L always → O) ✅ Monoalphabetic
- **Polyalphabetic** - Same letter can map to different substitutes
  - Example: HELLO → ABNZF (L → N and L → Z) ❌ NOT Monoalphabetic

2. Caesar Cipher (Shift Cipher)

- Each letter shifted by fixed number (key)
- Example: Key = 15, H (8th letter) + 15 = 23 → W

Quick Tip: A=1, B=2... Z=26, or remember positions: A=0, B=1... Z=25 for modular arithmetic

3. Transposition Cipher

- **Reorders (permutes)** symbols, doesn't replace them
- **Rail Fence Cipher** - Write in zigzag pattern across "rails", read row by row

4. Cryptanalysis Approaches

1. **Cryptanalytic attack** - Using mathematical weaknesses, patterns, frequencies
2. **Brute-force attack** - Trying every possible key

5. Letter Frequency (English - Most Common)

Most frequent: E, T, A, O, I, N, S, H, R

- E is the most common (~12.7%)
- T is second most common (~9.1%)

Common patterns to look for:

- "the" is most common 3-letter word
- If you see "ZWP" and guess ZW = th, then ZWP = "the"

6. Hash Functions

Property	Explanation
Fixed output	Any input length → fixed hash length
One-way	Cannot reverse hash to get original
Collision resistant	Different inputs shouldn't give same hash
MD5	128-bit digest (4 words of 32 bits)
SHA-1	160-bit digest (5 words of 32 bits)

Key Fact: Hash provides **integrity** (detects changes) but NOT authentication

Trick Question: Lossless compression is NOT a hash function because it's **reversible**

Lecture 03 - DES (up to Slide 12)

1. Basic Components (Remember what each does)

Component	Full Form	Function
XOR	Exclusive OR	Bitwise operation, reversible ($A \oplus B \oplus B = A$)
Rotation	-	Shifts bits left or right
P-Box	Permutation Box	Rearranges bits (like shuffling)
		- Straight: same number in/out
		- Compressed: fewer out than in

Component	Full Form	Function
		- Expanded: more out than in
S-Box	Substitution Box	Replaces input with different output (non-linear, provides security)

2. DES (Data Encryption Standard) Facts

- Created: **1977** by NBS (now NIST)
- Standard: FIPS PUB 46
- Block size: **64 bits**
- Key size: **56 bits** (effective - originally 64 with 8 parity bits)
- Status: Now **obsolete** (broken by brute force)

3. DES Operation Overview

1. **Initial Permutation (IP)** - Reorders 64-bit input
   - Even bits → Left half
   - Odd bits → Right half
2. **16 Rounds** of processing (Feistel structure)
   - Each round uses a different 48-bit subkey
3. **Final Permutation (FP)** - Inverse of IP

4. One DES Round (Feistel Function F)

```
text
Li = Ri-1
Ri = Li-1 ⊕ F(Ri-1, Ki)

Where F does:
1. Expand Ri-1 from 32 → 48 bits (Expansion P-box)
2. XOR with 48-bit subkey Ki
3. Pass through 8 S-boxes (each 6→4 bits) → 32 bits
4. Permute with 32-bit P-box
```

5. DES Decryption

- **Same algorithm** as encryption, but subkeys used in **reverse order**
- Round 1 uses SK16, Round 16 uses SK1

6. DES Security

- Key space: $2^{56} = 7.2 \times 10^{16}$ possible keys
- **Avalanche Effect** - Small change in plaintext/key → big change in ciphertext
- Broken by:
  - 1997: Internet distributed attack (months)
  - 1998: Dedicated hardware (days)
  - 1999: Combined (22 hours!)

Common Exam Traps to Avoid

Statement	Truth	Why?
"Algorithms must be secret"	❌ False	Algorithms are public, keys are secret
"Compression is a hash function"	❌ False	Compression is reversible, hash is one-way
"Hash provides authentication"	❌ False	Hash provides integrity only
"DES key is 64 bits"	⚠ Partially True	But only 56 bits are effective
"WPA2 is obsolete"	❌ False	WPA2 (AES) is still secure and widely used

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Quick Formula Sheet

1. Shift Cipher (Caesar)

```text
C = (P + K) mod 26
P = (C - K) mod 26
```

Where A=0, B=1, ... Z=25

2. Key Size & Brute Force Time

- Time doubles with each additional key bit
- 56-bit key: 2^{55} μs ≈ 1142 years (at 1 decryption/μs)
- 128-bit key: 2^{127} μs ≈ 5.4×10^{24} years

3. DES Round Count

- **DES**: 16 rounds
- **3DES**: 3 rounds of DES (encrypt-decrypt-encrypt with 2 or 3 keys)

Memory Tricks

Layers (A-T-N): "Apple Tastes Nice" → Application (PGP), Transport (SSL), Network (IPSec)

CIA Triad: "Can I Access?" → Confidentiality, Integrity, Availability

Attack Types: "PAM RMD" → Passive (Analysis, Message content), Active (Masquerade, Replay, Modification, DoS)

Hash Sizes: "MD5 = 128 (4×32), SHA-1 = 160 (5×32)" → MD5 has 4 words, SHA-1 has 5 words

DES Year: "DES born in 77, broken by 99" → 22 years of service

WiFi Order: "WEP WPA WPA2 WPA3" → 1,2,3 in order of security (WEP worst, WPA3 best)